When Half the Labels Are Missing: A Study on Facial Keypoints

PSL Practical Deep Learning, Group Project Report

1 Problem Formulation

Missing labels are the norm in real-world supervised learning. Datasets aggregated from heterogeneous sources rarely feature dense annotations. For instance, multi-task vision benchmarks often merge class-specific protocols[1], while crowdsourced collections reflect annotator-specific expertise[2]. Consequently, label missingness is typically dependent on the data distribution; this violates the Missing-At-Random (MAR) assumption and precludes standard imputation techniques[3].

Navigating this sparsity presents a fundamental tradeoff: discarding incomplete rows sacrifices statistical power, whereas omitting sparse columns restricts task breadth. While resolving this tradeoff is a recurring challenge across applied machine learning, evaluating novel methodologies on full-scale datasets introduces confounders that obscure underlying algorithmic mechanisms. Therefore, a controlled, smaller-scale experimental environment is essential to accurately isolate and evaluate the impact of specific design choices.

Facial keypoint detection as a testbed. We adopt the Kaggle Facial Keypoints Detection dataset as a controlled environment exhibiting this exact sparse-label structure. This multi-output regression task predicts up to 15 facial keypoints (30 real-valued pixel coordinates) from 96×96 grayscale images. The label matrix exemplifies the aforementioned missingness pattern: among 7,049 training examples, only 2,140 rows (30.4%) contain complete annotations. The remaining rows provide only four labeled keypoints (eye centers, nose tip, bottom lip center), leaving 26 coordinates unobserved. Consequently, 49.9% of all cell-level values are missing. Crucially, this missingness is structural: the 11 sparse keypoints are systematically co-observed or co-absent, reflecting an aggregation of distinct labeling protocols. This strictly precludes classical independent-MAR imputation, forcing an explicit methodological choice regarding data retention.

The row-keypoint tradeoff. Table 1 outlines three distinct strategies for handling the sparse matrix. A practitioner must prioritize either the number of keypoints, the number of training samples, or preserve both by adapting the loss function to accommodate missing values (NaNs). This inherent tension defines the sparse-label setting, as no single approach maximizes both dimensions.

Table 1: The row-keypoint tradeoff. The full label matrix has $7{,}049 \times 30 = 211.5$k coordinate cells but only 105.9k are non-NaN (49.9%). Each column is one way to present the data; no single column dominates in both row- and keypoint-count.

	Favour keypoints	Favour rows	Keep both
Rows (of 7,049)	2,140 (30.4%)	7,049 (100%)	7,049 (100%)
Keypoints (of 15)	15 (100%)	4 (27%)	15 (100%)
Usable coord. cells	64.2 k (30.4%)	28.2 k (13.3%)	**105.9 k (50.1%)**

The first two strategies represent standard listwise or column-wise deletion, sacrificing one dimension to guarantee dense data in the other. The third strategy is only viable if the training loss strictly ignores NaN cells. Whether the third column can actually dominate the first two in practice is what the methodological section 3.2 addresses; the rest of the study (augmentation, hyperparameter-tuning) conditions on the answer.

2 Model Design Choices

We compare three models in table 2: a two-layer MLP, a shallow CNN trained from random initialization, and a ResNet-18[4] initialized from ImageNet pretraining[1]. MLP-2 and CNN-Shallow are matched in parameter count (4.72M vs 4.75M) so any gap between them is attributable to the convolutional prior rather than to capacity. ResNet-18 further adds depth and pretrained features. All three share a 30-unit regression head.

Table 2: Three-model ladder.

Model	Architecture	Params	Prior
MLP-2	Flatten(96^2) $\rightarrow$ FC $9{,}216 \rightarrow 512 \rightarrow 30$	4.72M	None
CNN-Shallow	3 conv-pool stages 32/64/128 ch $\rightarrow$ FC $256 \rightarrow 30$	4.75M	Translation equiv.
ResNet-18 (pre.)	ImageNet-1k; conv1 $3 \rightarrow 1$ ch (RGB avg); fc $\rightarrow$ 30	11.2M	Hier. + IN feat.

3 Baseline

Section 1 posed the label-sparsity problem in general terms. The question we examine here is whether the row-keypoint tradeoff is unavoidable, or whether an appropriate choice of loss function lets a model supervise all rows and all keypoints simultaneously without injecting bias. To address this, this section will first establish naive baselines representing the two extremes of this tradeoff and subsequently formulate masked Mean Squared Error (MSE) as a principled resolution, rigorously evaluating whether it outperforms these extremes on held-out error.

3.1 Simple Dataset Split

The simplest resolution to the tradeoff identified in Section 1 is to strictly prioritize one dimension. We instantiate two baseline datasets representing the extremes: **data_small** prioritizes keypoint density by discarding any row containing NaNs (retaining 2,140 fully-labeled rows with 15 keypoints), while **data_large** prioritizes sample size by dropping the 11 sparse keypoints (retaining all 7,049 rows but supervising only the 4 baseline keypoints). Both yield a complete label matrix, allowing training via standard MSE, $\mathcal{L}_{\text{MSE}} = \frac{1}{Nd}\sum_{i,j}(\hat{y}_{ij} - y_{ij})^2$. However, both approaches incur a severe opportunity cost by discarding massive amounts of usable coordinate data.

Table 3: MSE loss across three models on both dataset splits.

Cohort	MLP-2	CNN	ResNet
data_small (15 kp)	0.00832	**0.00190**	0.00636
data_large (4 kp)	0.00792	**0.00328**	0.00774

As shown in Table 3, the CNN-Shallow architecture consistently outperforms both the MLP-2 and ResNet-18 baselines across both dataset splits. However, evaluating the relative efficacy of the two data-retention strategies (listwise versus column-wise deletion) remains impossible at this stage. Because plain MSE is averaged over a fundamentally different number of targets (15 keypoints versus 4 keypoints), the loss values across the two cohorts are mathematically incommensurable. Consequently, these naive baselines cannot yet reveal which strategy preserves more useful supervisory signal.

Before resolving this incommensurability, we define the foundational training protocol. To ensure all subsequent comparisons strictly isolate the effects of data retention and augmentation, every run in Sections 3 and 4 adheres to the fixed hyperparameters detailed in Table 4. The per-model learning rate is the single exception: ResNet-18 is trained at 10^{-4} because a higher rate destabilizes the pretrained features, while the from-scratch MLP-2 and CNN-Shallow use 10^{-3}.

Table 4: Training hyperparameters.

Hyperparameter	Value
Optimiser	Adam
Learning rate	10^{-3} (MLP/CNN); 10^{-4} (ResNet)
Weight decay	10^{-4}
LR schedule	Constant
Batch size	128
Epochs	30
Loss (small/large)	plain MSE
Loss (full)	masked MSE (§3.2)
Label norm.	$(y_{\text{px}} - 48)/48 \in [-1, 1]$
Model selection	best val MSE over 30 epochs
Random seed	42 (split, init, aug)
Hardware	1× NVIDIA A100 (40 GB)

3.2 Breaking the Trade-off

To break this tradeoff and utilize the full sparse matrix (data_full), we must adopt a strategy that retains every row and every keypoint while encoding missingness via a mask. Mask-based losses of this form are a standard device in partially-annotated multi-task learning[5,6]. We therefore replace plain MSE with **masked MSE**:

$$\mathcal{L}_{\text{masked}} = \frac{\sum_i \sum_{j=1}^{30} m_{ij} \cdot (\hat{y}_{ij} - y_{ij})^2}{\max\left(\sum_i \sum_{j=1}^{30} m_{ij},\, 1\right)}, \qquad m_{ij} \in \{0, 1\}.$$

Specifically, each coordinate is assigned a binary mask m_{ij}: $m_{ij} = 1$ if the label is present, 0 if NaN. Missing coordinates contribute 0 to both numerator and denominator, so they neither inject bias (unlike fillna(0)) nor cost supervision (unlike row-drop). By dividing strictly by the number of valid observations, the loss becomes a **per-observed-coordinate** average, rendering it structurally comparable across all three missingness regimes.

Under this formulation, on data_small, $m_{ij} = 1$ for all coordinates, so masked MSE reduces to plain MSE. On data_large, the entries corresponding to the 22 dropped columns are masked to zero. On data_full, the mask varies dynamically row-by-row. Ultimately, data_full retains all available cells. The results can be found in table 5 and figure 1.

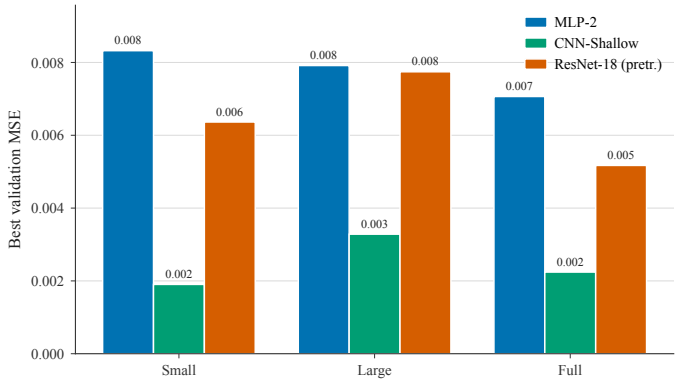


Figure 1: Full Baselines. Best-val MSE across the three label-handling strategies and three models, all on masked MSE. CNN-Shallow dominates every column; ResNet is penalized on the smallest cohort and recovers at scale.

Table 5: Mask-MSE loss across three models on full dataset.

Cohort	MLP-2	CNN	ResNet
data_small	0.00832	0.00190	0.00636
data_large	0.00792	0.00328	0.00774
data_full	**0.00706**	**0.00224**	**0.00517**

Findings. (1) Masked MSE does not universally dominate; for our best-performing model (CNN-Shallow), the naive listwise deletion baseline (data_small) surprisingly outperforms the maximally inclusive masked approach (data_full). This reveals that for architectures with strong spatial inductive biases, a smaller but perfectly dense label matrix provides a cleaner optimization landscape than a larger, sparsely masked alternative.

(2) CNN-Shallow universally dominates the leaderboard while ResNet suffers from over-parameterization in this regime. ResNet severely degrades on the task-restricted data_large baseline but achieves its best performance on data_full. This sensitivity demonstrates that high-capacity models rely heavily on the regularizing effect of maximum task breadth to prevent overfitting, rendering them vulnerable to column-dropping strategies.

4 Data Augmentation

Data augmentation serves as a standard regularizer in vision tasks[7,8], in our case we need to apply geometric transformations to both the image and its label. Certain transformations like horizontal flips can introduce structural inconsistencies on strongly yawed faces. We detail our augmentation pipeline below, comprising nine atomic operations, five compositional strategies, and a cohort-conditional gating mechanism to ensure anatomically valid transformations.

4.1 Atomic Transforms and Five Strategies

Our pipeline is constructed from nine atomic operations. Geometric operations jointly modify the image and its spatial coordinates: horizontal flip (G1, which incorporates a bilateral keypoint-pair swap to maintain anatomical consistency), rotation (G2), translation (G3), and scaling (G4). Photometric operations strictly alter pixel intensities without affecting coordinates: adjustments to brightness (P1) and contrast (P2), along with Gaussian noise (P3) and blur (P4). Finally, E1 introduces keypoint-aware random erasing. Table 6 outlines the five compositional strategies, while Figure 2 visualizes their empirical effects.



Figure 2: Visual effect of each strategy on one face. Rows are S1 to S5; columns are four independent random draws. S1 keeps geometry; S2 to S5 apply geometric perturbation; S5 adds erasing patches.

Table 6: The five augmentation strategies.

ID	Composition	What it tests
S1	P1+P2+P3	Photometric only, conservative baseline.
S2	G1+G2+G3	Geometric only: does spatial perturbation alone regularize?
S3	S1+S2	Photometric + geometric stacked, the standard recipe.
S4	S3+G4+P4	S3 plus scale and blur, heavier dose.
S5	S3+E1	S3 plus keypoint-aware random erasing.

4.2 Cohort Identification and Geometric-atom Gate

Geometric transformations are not universally safe across diverse facial poses. For instance, a 60°-yaw (side-view) face has keypoints that sit near the silhouette; applying a horizontal flip (hflip) or a large rotation forces these points into anatomically impossible positions, corrupting the label matrix. To mitigate this, we classify each training instance into one of three distinct cohorts based on inter-eye geometry, subsequently gating the geometric atoms per cohort (Table 7, Figure 3). The routing thresholds were mainly set via Exploratory Data Analysis (EDA) on the densely annotated data_small subset and applied to the two larger cohorts.

Table 7: Cohort identification rule and geometric-atom gate. dx: horizontal inter-eye distance; $|dy|$: vertical offset; p_{G1}: hflip probability; $\theta_{G2}^{\max}$: rotation cap. Photometric atoms fire uniformly.

Cohort	Rule	p_{G1}	$\theta_{G2}^{\max}$
Frontal	$dx \geq 25.45, \lvert dy\rvert < 3$	0.5	12°
Asym.	$dx \geq 25.45, \lvert dy\rvert \geq 3$	0.1	12°
Hard-prof.	$dx < 25.45$	0.0	5°

Figure 3 shows the rule on three exemplar faces from data_full: a symmetric frontal face (hflip allowed at $p = 0.5$), a tilted but still mostly symmetric face (hflip rare, $p = 0.1$), and a yawed profile (hflip disabled, rotation capped at 5°). The same rule is applied to every training sample.

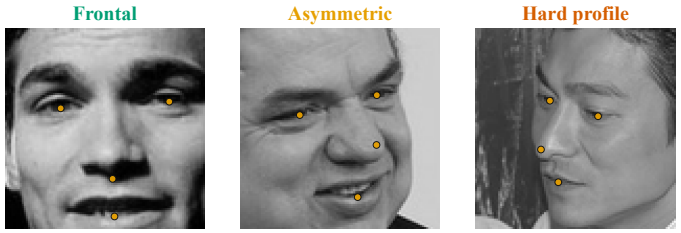


Figure 3: Cohort identification on three exemplar data_full faces. The inter-eye dx / $|dy|$ rule routes each sample to frontal / asymmetric / hard-profile, and the per-cohort gate determines which geometric atoms fire.

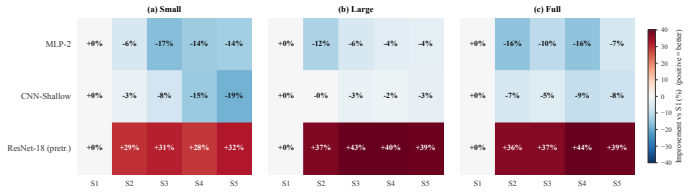


Figure 4: Augmentation benefit: percent *improvement* in val MSE relative to S1 (positive means augmentation helps, shown in red; negative means augmentation hurts, shown in blue). Three strata are visible: ResNet benefits on every cohort; CNN is essentially flat; MLP is hurt by geometric strategies.

4.3 Results

Figure 4 visualizes each cell's improvement from S1, and Table 8 lists the winning strategy per cell.

Table 8: Part 2 best-strategy-per-cell summary. Best val MSE across S1–S5; improvement % vs S1. ResNet wins on the two larger cohorts; CNN on the smallest.

Cohort	Model	Best	MSE	Δ vs S1
data_small	MLP	S1	0.00816	0.0%
	CNN	S1	**0.00189**	0.0%
	ResNet	**S5**	0.00257	**+31.9%**
data_large	MLP	S1	0.00794	0.0%
	CNN	S1	0.00337	0.0%
	ResNet	**S3**	**0.00298**	**+42.8%**
data_full	MLP	S1	0.00664	0.0%
	CNN	S1	0.00218	0.0%
	ResNet	**S4**	**0.00198**	**+43.7%**

5 Hyperparameter Tuning and Final Evaluation

Sections 3.2 and 4 maintained fixed training hyperparameters (Table 4) to ensure pairwise comparisons strictly isolated individual factors. Here, we relax this constraint for the optimal configuration (data_full trained with masked MSE) and, for the first time in this study, report final performance on the held-out test set.

5.1 Search protocol

For each architecture, we conduct an 8-trial random search over key training hyperparameters, training for up to 40 epochs with early stopping. For MLP-2 and CNN-Shallow, the search space encompasses learning rate, batch size, weight decay, and hidden-layer width (for the MLP). For ResNet-18, we additionally tune two augmentation parameters: the application probability p and the scaling range $[s_{\text{lo}}, s_{\text{hi}}]$. This aligns with our earlier observation that augmentation primarily benefits the high-capacity pretrained model. Finally, the optimal trial for each model (selected

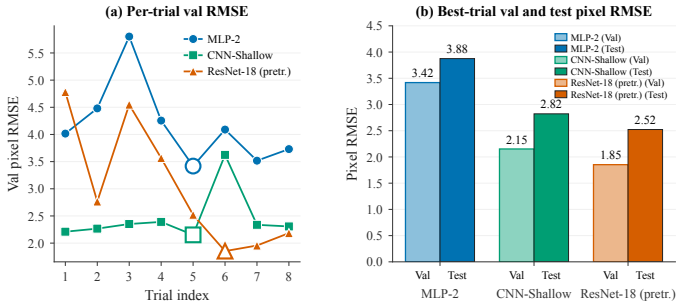


Figure 5: Hyperparameter search on data_full. (a) Val pixel RMSE for all 8 trials per model; white circles mark each model's best-of-8. (b) Best-trial val and test pixel RMSE side-by-side; val-to-test gap is uniform across models (approximately 0.4–0.7 px), so the val-selected ranking holds on test.

via validation MSE) is evaluated precisely once on the test split, which remained strictly isolated throughout all prior exploratory experiments.

5.2 Results

Table 9 reports each model's best-of-8 trial with its selected hyperparameters and val/test pixel RMSE.

Table 9: Each row is the best-of-8 trial for one model on data_full. *Val px* and *test px* are pixel RMSE at the best epoch.

Model	lr	bs	Val MSE	Val px	Test px
MLP-2	4.3e-4	64	0.00507	3.42	3.88
CNN-Shallow	5.2e-4	128	0.00201	2.16	2.82
ResNet-18	4.2e-4	64	**0.00149**	**1.85**	**2.52**

Figure 5 visualizes the search trajectory and the val-vs-test gap per model.

Hyperparameter optimization reduces the ResNet-18 validation MSE from 0.00395 to 0.00149, a 62% improvement. This substantial gain propels ResNet-18 ahead of CNN-Shallow across both validation and test metrics. Furthermore, the generalization gap (validation-to-test RMSE difference) remains remarkably consistent across all three architectures (0.46 px for MLP, 0.66 px for CNN, and 0.67 px for ResNet). Consequently, the model ranking established during validation is perfectly preserved on the unseen test split. Achieving a test pixel RMSE of **2.52 px on a 96×96 image**, this optimal ResNet-18 configuration represents the definitive generalization benchmark of this study.

6 Findings & Limitations

6.1 Findings

The experiments yield three distinct methodological insights.

Finding 1: Matrix density versus data volume. While masked MSE successfully standardizes the optimization scale across missingness patterns, retaining the full sparse matrix (data_full) does not universally dominate naive deletion strategies. For architectures with strong spatial inductive biases (CNN-Shallow), listwise deletion (data_small) actually yields superior validation MSE, demonstrating that a smaller, perfectly dense label matrix can outcompete sparsely masked data volume. Conversely, the high-capacity ResNet-18 achieves its best baseline performance on data_full, confirming that overparameterized models rely heavily on the regularizing effect of maximum task breadth to prevent overfitting.

Finding 2: Augmentation returns are capacity- and architecture-gated. Augmentation is not a universal regularizer. We observe three distinct regimes: ResNet-18 sees strong gains (up to +43.7%) as geometric perturbations close its memorization gap. CNN-Shallow sees marginal impact, as its native convolutional prior provides sufficient regularization. MLP-2 suffers severe degradation (-16%), as geometric transformations inject destructive noise without a spatial prior. This stratification proves that fair evaluation necessitates optimal, per-model augmentation regimes.

Finding 3: Validation-selected rankings transfer reliably to the test domain. The model hierarchy established during validation is rigorously preserved on the held-out test split. Following hyperparameter optimization, ResNet-18 achieves a test pixel RMSE of 2.52, strictly outperforming CNN-Shallow and MLP-2. The uniform generalization gap across architectures confirms the stability of the evaluation protocol. Furthermore, ResNet-18's 62% reduction in validation MSE post-tuning indicates that default learning rates severely bottlenecked the pretrained model in earlier phases, reinforcing the capacity dynamics identified in Finding 2.

6.2 Limitations

(i) **Confounded Zero-Dose Baseline:** The S1 augmentation strategy serves as the baseline for Section 4, but it introduces photometric noise rather than being a strict zero-augmentation control. Consequently, the calculated marginal utility of geometric transformations (S2–S5) is measured against an already-regularized baseline, potentially understating the absolute regularizing impact of spatial augmentation on high-capacity models.

(ii) **Suboptimal Hard-Boundary Gating:** The geometric safety thresholds (dx, $|dy|$) enforce rigid, deterministic boundaries for transformation caps. This ignores the continuous nature of 3D facial rotation; a marginally sub-threshold face receives a full rotation allowance. This artificial bifurcation may cap the regularizing potential for borderline instances.

References

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